

United Kingdom 2015

1 Survey Description

Survey: Understanding Society (2015)

Link to the document: <https://www.understandingsociety.ac.uk/about/british-household-panel-survey/>

Sample: Understanding Society is a longitudinal, household panel, multi-scope study conducted by the Institute for Social and Economic Research at the University of Essex. The Study began in 2009 as a continuation of the British Household Panel Study (BHPS), which ran from 1991 to 2008. The Understanding Society sample design consists of four components: the General Population Sample (GPS), the Ethnic Minority Boost Sample (EMBS), the former British Household Panel Survey (BHPS) sample, and the Immigrant and Ethnic Minority Boost Sample (IEMBS). Understanding Society employs a probability sampling approach with a complex sample design, incorporating clustering and stratification, and includes unequal selection probabilities to ensure representation of population subgroups. The GPS uses a proportionately stratified, clustered sample of residential addresses from the Postcode Address File in England, Scotland, and Wales, and an unclustered systematic random sample from the Land and Property Services Agency list in Northern Ireland.. There are 39,226 individuals in the total sample and 11,796 individuals in the analysis sample. Section 3 of this document describes the prevalence and pattern of missing data.

Weights: Understanding Society employs a probability sampling approach with a complex sample design, incorporating clustering and stratification, and includes unequal selection probabilities to ensure representation of population subgroups. The GPS uses a proportionately stratified, clustered sample of residential addresses from the Postcode Address File in England, Scotland, and Wales, and an unclustered systematic random sample from the Land and Property Services Agency list in Northern Ireland. The weights account for this stratification.

Outcome: The outcome variables are annual equivalized household disposable total income in dollars PPP 2017.¹

Circumstances:

- Race/Ethnicity (described in Table 1)
- Sex (female, male)
- Country of Birth (24 categories described in Table 1)
- Fathers's edu. (4 levels of education, described in Table 2)
- Mother's edu. (4 levels of education, described in Table 2)
- Father's occ. (8 categories NSSEC, described in Table 3)
- Mother's occ. (8 categories NSSEC, described in Table 3)

¹Income variable was equivalized using the square root scale.

2 Descriptive Statistics

Table 1: Respondant's socio-demographics - 2015

	Analysis sample	Total sample
	(N=11,796)	(N=39,226)
Gender		
0 Male	4,932 (41.8%)	17,431 (44.4%)
1 Female	6,864 (58.2%)	21,795 (55.6%)
Ethnicity		
1 Black	507 (4.3%)	1,883 (4.8%)
2 White	10,514 (89.1%)	31,491 (80.3%)
3 Asian	541 (4.6%)	4,446 (11.3%)
4 Mixed	181 (1.5%)	717 (1.8%)
6 Other	53 (0.4%)	332 (0.8%)
Missing	0 (0%)	357 (0.9%)

Table 2: Parental education - 2015

	Analysis sample	Total sample
	(N=11,796)	(N=39,226)
Mother's education (levels)		
1 Primary	4,590 (38.9%)	13,420 (34.2%)
2 Secondary Lower	3,582 (30.4%)	7,794 (19.9%)
3 Secondary Upper	2,430 (20.6%)	4,615 (11.8%)
4 Tertiary	1,194 (10.1%)	2,055 (5.2%)
Missing	0 (0%)	11,342 (28.9%)
Father's education (levels)		
1 Primary	4,477 (38.0%)	11,328 (28.9%)
2 Secondary Lower	2,626 (22.3%)	5,771 (14.7%)
3 Secondary Upper	3,210 (27.2%)	6,655 (17.0%)
4 Tertiary	1,483 (12.6%)	3,071 (7.8%)
Missing	0 (0%)	12,401 (31.6%)

Table 3: Parental occupation - 2015

	Analysis sample	Total sample
	(N=11,796)	(N=39,226)
Mother's occupation		
1 Managers	552 (4.7%)	1,430 (3.6%)
2 High Professional	911 (7.7%)	2,104 (5.4%)
3 Low Professional	1,896 (16.1%)	4,287 (10.9%)
4 Intermediate	1,202 (10.2%)	2,719 (6.9%)
5 Seftemployed	1,726 (14.6%)	4,811 (12.3%)
6 Technical	1,835 (15.6%)	4,197 (10.7%)
7 Semi routine	1,578 (13.4%)	3,844 (9.8%)
8 Routine	2,096 (17.8%)	5,467 (13.9%)
Missing	0 (0%)	10,367 (26.4%)
Father's occupation		
1 Managers	122 (1.0%)	188 (0.5%)
2 High Professional	215 (1.8%)	341 (0.9%)
3 Low Professional	2,413 (20.5%)	3,605 (9.2%)
4 Intermediate	2,253 (19.1%)	3,413 (8.7%)
5 Seftemployed	776 (6.6%)	1,246 (3.2%)
6 Technical	265 (2.2%)	383 (1.0%)
7 Semi routine	3,030 (25.7%)	4,961 (12.6%)
8 Routine	2,722 (23.1%)	4,502 (11.5%)
Missing	0 (0%)	20,587 (52.5%)

Table 4: Respondant's income - 2015

	N	Mean	SD	Median	Min	Max	Missing
Analysis sample	11,796	55,613	45,057	46,979	18.494	1,328,570	0
Total sample	39,226	52,192	82,441	43,185	5.899	11,550,107	296

3 Missing data analysis

3.1 Missing patterns

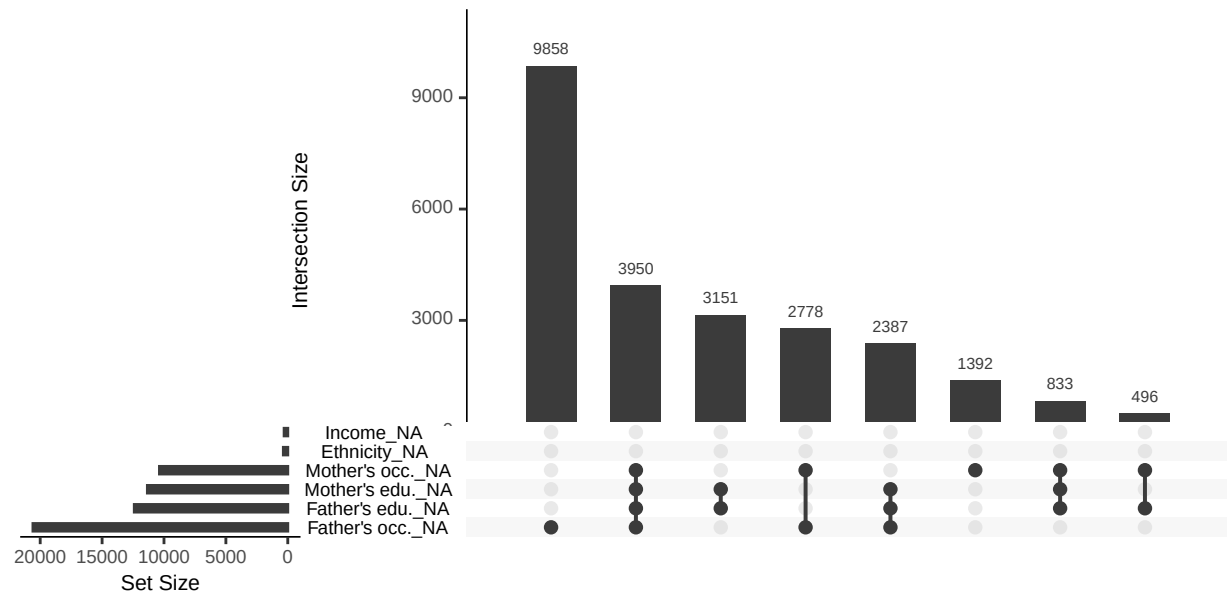


Figure 1: Missing patterns: *Left*: Marginal distribution of missing observations per variable. *Right*: Combination of missingness across cases

3.2 Differences in expected total equivalized household income between samples

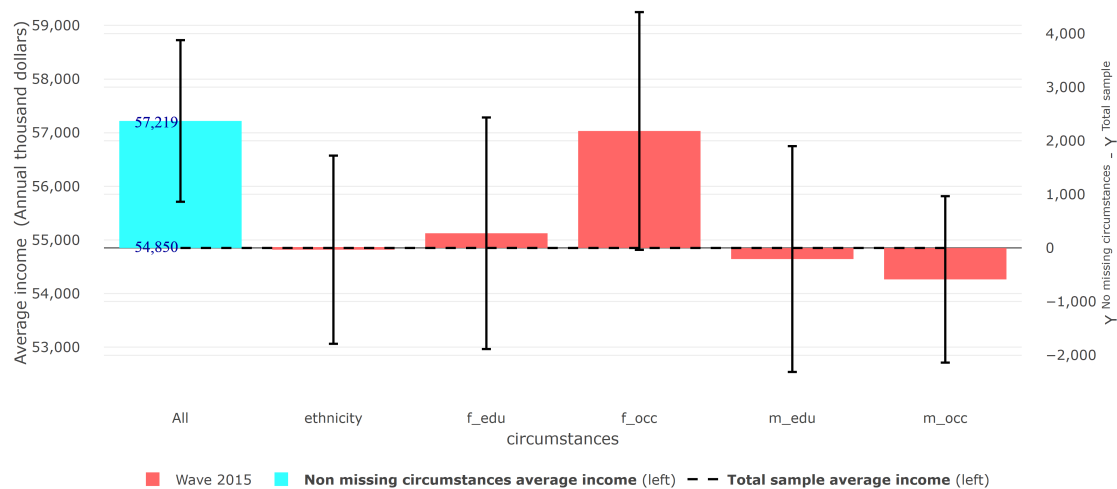


Figure 2: Differences in expected total equivalized household income between the sample with non-missing circumstances and the total sample

3.3 Gini coefficient

Table 5: Gini coefficient in analysis sample and total sample

Wave	Sample	Gini	Lower bound	Upper bound	Average income
Wave 2015	Analysis sample	0.342	0.329	0.348	57,219
Wave 2015	Total sample	0.353	0.342	0.375	54,850

3.4 Differences in Gini coefficient between samples

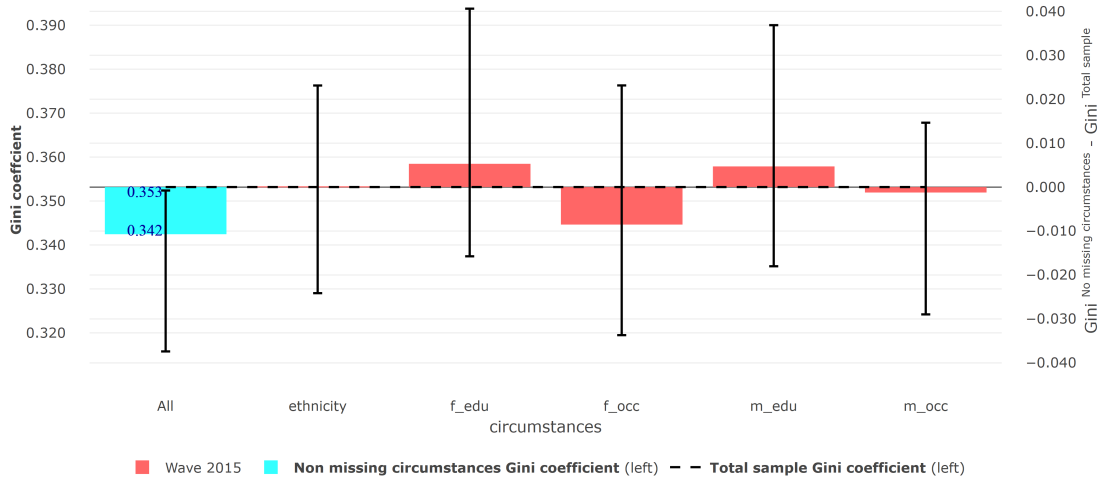


Figure 3: Differences in Gini coefficient between the sample with non-missing circumstances and the total sample